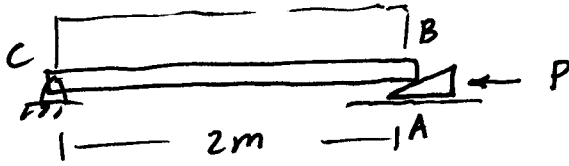


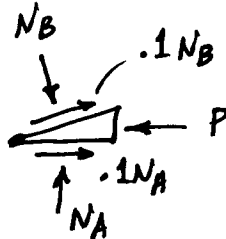
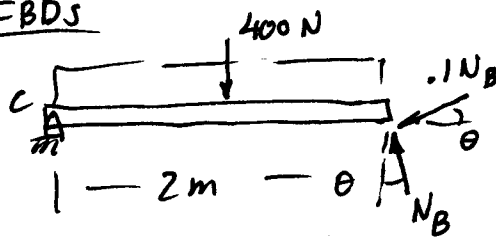
Wedge Raising Case : P needed to raise beam

200 N/m

For wedge, let $\mu = .1$, $\theta = 10^\circ$.



FBD's



"Push-In" Case,
frictions act
opposite of
relative
motion.

Beam :

$$+\uparrow \sum M_C = 0; \quad 400(1m) + .1N_B(2m)(\sin \theta) - N_B \cos \theta(2m) = 0$$

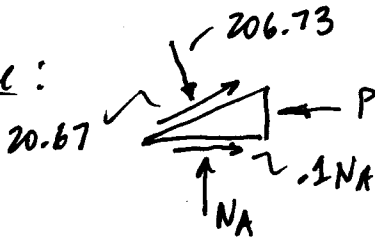
$$400 = N_B(2 \cos \theta^{10^\circ} - .2 \sin \theta^{10^\circ})$$

$$= N_B(1.9349)$$

$$\boxed{N_B = 206.73 \text{ N}}$$

$$\boxed{.1N_B = 20.67 \text{ N}}$$

Wedge :



$$+\uparrow \sum F_y = 0; \quad N_A - 206.73 \cos 10 + 20.67 \sin 10 = 0;$$

$$\boxed{N_A = 200 \text{ N}} \text{ as expected}$$

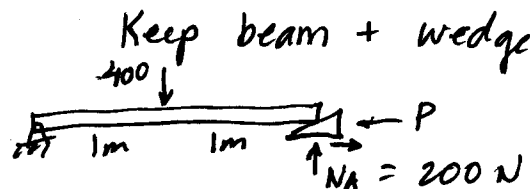
$$\boxed{.1N_A = 20 \text{ N}}$$

$$+\leftarrow P - .1N_A - 206.73 \sin 10 - 20.67 \cos 10 = 0$$

$$\boxed{P = 76.25 \text{ N} \leftarrow}$$

Why is $N_A = 200 \text{ N}$?

same FBD:



Keep beam + wedge together on

Not true for other cases...

